



Experimental and kinetic modeling study of benzyl alcohol pyrolysis

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ABSTRACT

The pyrolysis of benzyl alcohol (A1CH₂OH) at 30 and 760 torr was studied in a flow tube reactor. The synchrotron radiation photoionization and molecular-beam mass spectrometry (MBMS) techniques were used to identify and quantify 35 intermediates and products including some small species, several monophenyl ring species, and a large number of polyaromatic hydrocarbons (PAHs). A comprehensive chemical kinetic reaction model involving 376 species and 2171 reactions was developed with reasonable prediction. According to the rate-of-production analysis, the consumption of A1CH₂OH is mainly proceeded with the H-abstraction reactions with H atoms and OH radicals under both pressures. H-abstractions and unimolecular dissociation dehydrogenation reactions also play important roles in the subsequent formation of PAHs. Phenyl radicals, benzene (A1), and benzyl (A1CH₂) radicals are important intermediates in the pyrolysis of A1CH₂OH, which provide various ways for the formation of PAHs. Sensitivity analysis presents that the most significant sensitive reaction of A1CH₂OH consumption is the unimolecular initiation reaction by the C–O bond breaking from A1CH₂OH under both pressures. The combination of A1CH₂ radicals with H atoms to form toluene (A1CH₃) has the most inhibiting influence on the A1CH₂OH consumption under both pressures, while it has little sensitive effects for the oxidation process of A1CH₂OH consumption. The comparison of the A1CH₂OH pyrolysis and oxidation results show that the peak mole fractions of A1 increase, while the peak mole fractions of phenol (A1OH) decrease with the increase of equivalence ratio (ϕ) under the same pressure. It should be noted that the peak mole fractions of benzaldehyde (A1CHO) are relatively stable with various conditions. These results provide a theoretical basis for further study of the combustion chemical kinetics of A1CH₂OH and its application in bio alternative fuel.

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1. Introduction

Biofuels have played important roles in dealing with global energy shortage and greenhouse gas emissions in recent years. The International Energy Agency proposed to increase the share of biofuels in transportation fuels from 2% in 2011 to 27% by 2050 [1]. 2.1 billion tons of CO₂ emissions will be reduced every year once the goal is achieved [1,2]. Lignin, as one of the three major polymers of biomass, accounts for about 30% of the total biomass [3,4]. At the same time, as the only natural polymer with repetitive aromatic structure unit in nature, lignin has great potential application value. Anisole (A1OCH₃), benzyl alcohol (A1CH₂OH), and other biological alternative fuels can be derived from lignin pyrolysis [5].

These oxygenated aromatic biological fuels have attracted great attentions due to their low reactivity and emission mitigating, which are interesting alternative fuels suitable for the concept of low temperature combustion.

A1CH₂OH and A1OCH₃ are isomers, and they are also relatively simple potential biofuels of C₇ oxygenated aromatics. However, most of the oxygenated aromatic biofuels combustion studies focused on A1OCH₃. Numerous investigations of A1OCH₃ were presented regarding its flame [6–8], oxidation [9–11], pyrolysis [8,10,12–15] and kinetic models [8,10,13,15]. Compared with A1OCH₃, the combustion study of A1CH₂OH still remains limited although they have similar properties. The reported results of oxygenated aromatic fuels from Zhou et al. [16–18] showed that A1CH₂OH could significantly improve engine efficiency with high performance and reduce the particulate matter emissions in the combustion process of blended biodiesel fuel. Frisch et al. [19] studied the thermal decomposition process of A1CH₂OH in the temperature range of 1200–1600 K. Husson et al. [20] reported that

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$\text{A1CH}_2\text{OH}$ was dissociated to form A1CHOH radicals during the oxidation process of ethylbenzene. Herbinet et al. [21] developed a oxidation mechanism of benzene and toluene by considering some reactions of $\text{A1CH}_2\text{OH}$. Narayanaswamy et al. [22] reported the mechanism of aromatic compounds, including the O–H bond cleavage of $\text{A1CH}_2\text{OH}$ and the subsequent reactions. The above study provided some sub-mechanisms for the combustion kinetic study of $\text{A1CH}_2\text{OH}$ while these sub-mechanisms had not been validated due to the absence of experimental data and accurate rate constants. Recently, our group reported the oxidation behavior of $\text{A1CH}_2\text{OH}$ [23] and played role in the development of a new kinetic model along with the theoretical calculation. However, it is still uncertain whether this model can reasonably simulate the effect of pressure on $\text{A1CH}_2\text{OH}$ combustion due to the shortage of experimental data related to variable pressure. Therefore, the model requires additionally validation against the experimental data of $\text{A1CH}_2\text{OH}$ pyrolysis and verify its prediction performance.

In this work, pyrolysis of $\text{A1CH}_2\text{OH}$ was studied at 30 and 760 torrs to obtain a deeper understanding of the combustion chemistry of $\text{A1CH}_2\text{OH}$. The synchrotron radiation photoionization and molecular-beam mass spectrometry (MBMS) techniques were used to identify and quantify the species in the pyrolysis process. A detailed kinetic model was further developed based on the previous study used for $\text{A1CH}_2\text{OH}$ oxidation. It was updated to predict the species profiles obtained through pyrolysis and oxidation of $\text{A1CH}_2\text{OH}$. $\text{A1CH}_2\text{OH}$ pyrolysis was analyzed through sensitivity and rate-of-production (ROP) analysis with respect to its consumption routes, significant promoting and restraining of pathways. The verified model can be used for $\text{A1CH}_2\text{OH}$ as a potential bio-alternative fuel, which is contributing to further understand the combustion characteristics of $\text{A1CH}_2\text{OH}$ in diesel and aviation oil.

2. Experimental method

The pyrolysis experiments of $\text{A1CH}_2\text{OH}$ were performed in the National Synchrotron Radiation Laboratory (NSRL) using the BL03U beamline. More detailed information about the instrument and equipment has been presented somewhere else [24–28], and only concise description is given here. The experimental setup consists of four sections: a homemade reflection time of flight mass spectrometer (RTOF-MS), a photoionization chamber with a molecular beam sampling system, a differential pump chamber and a pyrolysis chamber with an electric furnace and flow tube. The corundum flow tube reactor is made of α -alumina with an outer diameter of 10 mm, an inner diameter of 7 mm and a length of 400 mm. The temperature distribution along the axis of the flow tube was measured by the S-type thermocouple, as shown in Fig. 1. The uncertainty of the measured temperature was estimated to be ± 35 K. Detailed temperature data are shown in the supplemental material (SM). Argon (Ar) and krypton (Kr) were purchased from Nanjing Special Gas Inc., (purity $\geq 99.999\%$). $\text{A1CH}_2\text{OH}$ was purchased from Aladdin Reagent Co., Ltd., Shanghai, China, (purity $\geq 99.5\%$). The high-pressure infusion pump (FL2200, China Fuli Co., Ltd.) was used to control the feeding rate of $\text{A1CH}_2\text{OH}$ and $\text{A1CH}_2\text{OH}$ was evaporated in the vaporizer at 523 K. The total flow rate was 1000 sccm, consisting 985 sccm of Ar, 10 sccm of Kr for calibration, and 5 sccm of $\text{A1CH}_2\text{OH}$. The MKS mass flow controllers were used to control the flow rates of Ar and Kr. The detailed experimental conditions for the pyrolysis of $\text{A1CH}_2\text{OH}$ are presented in Table 1. The pyrolysis products were sampled by quartz nozzle. The diameters of the nozzle were 350 μm at 30 torr and 75 μm at 760 torr respectively. The samples were then ionized by vacuum ultraviolet light. The ions generated were collected and analyzed by the microchannel plate after entering RTOF-MS whose detection sensitivity is 1 ppm.

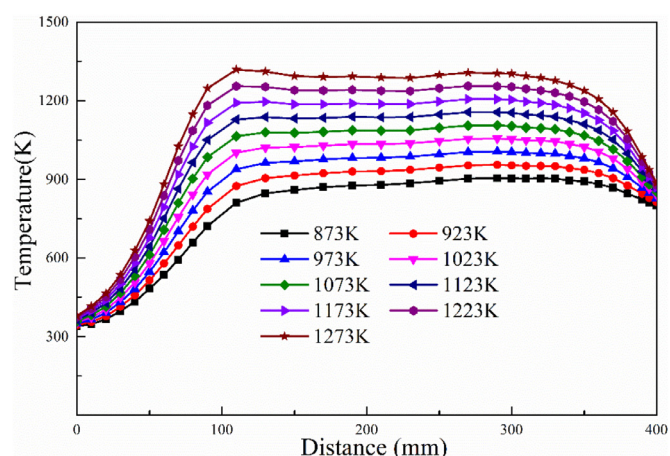


Fig. 1. The temperature inside the flow tube varies with distance, 0 mm is the inlet temperature and 400 mm is the temperature at the sampling point.

Table 1
Detailed experimental conditions for pyrolysis of $\text{A1CH}_2\text{OH}$.

Pressure (Torr)	Temperature (K)	Mole fraction (%)		
30.0	900–1275	$\text{A1CH}_2\text{OH}$	Kr	Ar
		0.5	1.0	98.5
760.0	875–1223	$\text{A1CH}_2\text{OH}$	Kr	Ar
		0.5	1.0	98.5

The pyrolysis experiments of $\text{A1CH}_2\text{OH}$ were carried out in two modes. One was to record the mass spectra with the temperature at specific photon energies and obtain the mole fraction profiles. The selected photon energies are presented in Tables S1–2 (SM). Another mode was to scan the mass spectra of intermediates and products under different ionization energies (IEs) to get the photoionization efficiency (PIE) spectra for qualitative analysis of unknown products during pyrolysis. The scanning temperature and selected photon energy range along with other conditions are shown in Table 2. Methods for the identification of intermediates and the calculation of mole fraction have been reported elsewhere [28–30]. The photoionization cross sections (PICSS) of most species are obtained from the online data of the photoionization mass spectrometry center of NSRL [31]. The selected PICSS data for the identified species are shown in Table S3 (SM). For the major species, the uncertainties of the obtained mole fractions are within ± 5 –10% [32,33]. The uncertainties are in the range of $\pm 25\%$ [34,35] for the intermediates with known PICSS. The uncertainty factors are evaluated to be two [35] for the species with speculative PICSS.

3. Modeling

The modeling for $\text{A1CH}_2\text{OH}$ pyrolysis was carried out in the Plug Flow Reactor (PFR) code from the Chemkin-Pro Pro-software [36]. Based on our recent models of A1CHO [37] and $\text{A1CH}_2\text{OH}$ oxidation [23], a new kinetic model involving 376 species and 2171 reactions was developed. Based on the reaction mechanism of alcohols [38–41], the H-abstraction reactions and unimolecular initiation reactions in the primary mechanism of $\text{A1CH}_2\text{OH}$ were updated. The thermochemical data were taken from the A1CHO mechanism [37,42] and $\text{A1CH}_2\text{OH}$ oxidation mechanism [23]. The critical promoting and inhibiting reaction routes of $\text{A1CH}_2\text{OH}$, during the consumption and intermediates formation were obtained through sensitivity and ROP analysis.

Table 3 lists the rate constants of some critical reactions in the $\text{A1CH}_2\text{OH}$ primary mechanism. The unimolecular initiation reactions of $\text{A1CH}_2\text{OH}$ (R1–R5) can generate A1CHOH radicals, phenyl

Table 2
Detail photon energy scan conditions for PIE spectra.

Pressure (torr)	Photon energy (eV)	Temperature (K)	Step width (eV)	Time (s)	Undulator offset	Repeat
30.0	7.0–10.0	1075	0.03	120	0	1
760.0	7.0–10.0	1098	0.03	120	0	1

Table 3

A list of selected reactions in the present A1CH₂OH primary mechanism, units are s⁻¹, cm³, and cal/mol.

No.	Reactions	A	N	Ea	References
<i>Unimolecular initiation reactions</i>					
R1	A1CHOH+H<=>A1CH ₂ OH	4.00E+13	0.00	0.00	[38] ^a
R2	A1CH ₂ OH=A1-+CH ₂ OH	1.95E+27	-3.16	107,447.00	[37]
R3	A1CH ₂ OH=A1CHO+H ₂	7.24E+11	0.10	91,010.00	[37] ^b
R4	A1CH ₂ +OH<=>A1CH ₂ OH	1.50E+13	0.00	0.00	
R5	A1CH ₂ O+H(+M)=A1CH ₂ OH(+M) LOW/ 4.660e+41 -7.440 14,079.83 / TROE/0.7 100.00 90,000.00 10,000.00 / H ₂ /2.00/ H ₂ O/12.00/ CO ₂ /3.60/ CO/1.75/ AR/0.70/ CH ₄ /2.00/ C ₂ H ₆ /3.00/	2.43E+12	0.52	50.19	[22]
<i>H-abstraction reactions</i>					
R6	A1CH ₂ OH+H=A1CH ₂ O+H ₂	9.45E+02	3.14	8701.00	[39] ^c
R7	A1CH ₂ OH+H=A1CHOH+H ₂	8.80E+04	2.70	2915.00	[40] ^d
R8	A1CH ₂ OH+O=A1CH ₂ O+OH REV/ 4.99E-05 4.891 -9.750E+02/	1.46E-03	4.73	1727.00	[41] ^e
R9	A1CH ₂ OH+O=A1CHOH+OH REV/ 1.69E+02 2.899 7.704E+03/	1.45E+05	2.47	8760.00	[41] ^e
R10	A1CH ₂ OH+CH ₃ =A1CH ₂ O+CH ₄	1.02E+00	3.57	8221.00	[41] ^e
R11	A1CH ₂ OH+CH ₃ =A1CHOH+CH ₄	1.99E+01	3.37	7634.00	[41] ^e
R12	A1CH ₂ OH+HO ₂ =A1CH ₂ O+H ₂ O ₂	6.47E-07	5.30	10,530.00	[41] ^e
R13	A1CH ₂ OH+HO ₂ =A1CHOH+H ₂ O ₂	3.50E-05	5.26	8268.00	[41] ^e
R14	A1CH ₂ OH+HCO=A1CH ₂ O+CH ₂ O	3.40E+04	2.50	13,500.00	[41] ^e
R15	A1CH ₂ OH+HCO=A1CHOH+CH ₂ O	1.00E+07	1.90	17,000.00	[41] ^e
R16	A1CH ₂ OH+CH ₂ OH=A1CH ₂ O+CH ₃ OH	1.20E+02	2.76	10,800.00	[41] ^e
R17	A1CH ₂ OH+CH ₂ OH=A1CHOH+CH ₃ OH	6.00E+01	2.95	12,000.00	[41] ^e
R18	A1CH ₂ OH+CH ₃ O=A1CH ₂ O+CH ₃ OH	2.30E+10	0.00	2900.00	[41] ^e
R19	A1CH ₂ OH+CH ₃ O=A1CHOH+CH ₃ OH	1.50E+11	0.00	4500.00	[41] ^e
R20	A1CH ₂ OH+C ₂ H ₅ =A1CH ₂ O+C ₂ H ₆	1.00E+11	0.00	9200.00	[41] ^e
R21	A1CH ₂ OH+C ₂ H ₅ =A1CHOH+C ₂ H ₆	2.00E+11	0.00	11,000.00	[41] ^e
R22	A1CH ₂ OH+OH=A1CH ₂ O+H ₂ O REV/ 9.038E+09 0.896 1.612E+04 /	1.50E+10	0.80	2534.00	[38] ^f
R23	A1CH ₂ OH+OH=A1CHOH+H ₂ O REV/ 1.324E+09 0.843 2.276E+04 /	5.56E+10	0.50	-380.00	[38] ^f
R24	A1CH ₂ OH+O ₂ =A1CH ₂ O+HO ₂ REV/ 2.980E+12 -0.146 -7.791E+03 /	1.00E+13	0.00	48,200.00	[38] ^f
R25	A1CH ₂ OH+O ₂ =HO ₂ +A1CHOH REV/ 2.356E+11 0.101 3.021E+03 /	2.00E+13	0.00	49,460.00	[38] ^f
<i>Ipsso-addition reactions</i>					
R26	A1CH ₂ OH+OH=A1OH+CH ₂ OH	7.83E+02	2.88	3221.80	[37]
R27	A1CH ₂ OH+H=A1+CH ₂ OH	2.40E+13	0.00	5120.00	[37]
R28	A1CH ₂ OH+CH ₃ =A1CH ₃ +CH ₂ OH	2.00E+00	2.80	2326.40	[23]

Note: ^a Referred to the similar reaction of *i*-propanol; ^b Estimated from the radical combination reaction; ^{c, e, f} Referred to the homologous reaction of *n*-butanol; ^d Referred to the analogous reaction of *i*-butanol.

radicals, A1CHO, A1CH₂ radicals and A1CH₂O radicals. The rate coefficient of R1 was referred to as a similar reaction from *i*-propanol [38]. The rate coefficients of R2 and R3 were taken from Chen et al. [37]. The rate coefficient of R4 was adopted from the radical combination reaction. The pressure-dependent rate constants of the O–H bond-breaking reaction (R5) was obtained from Narayanaswamy et al. [22]. Comparison of the revised rate constants of R1 and R4 with those from the model of Zhou et al. [11] are shown in Fig. S1(SM). As an important reaction part for the consumption of A1CH₂OH, H-abstraction reactions occupy a large proportion at the primary mechanism. Two H-abstraction sites on the side chain of A1CH₂OH show that it can generate A1CHOH and A1CH₂O radicals. R6 and R7 were the H-abstraction reactions with H atoms in which the rate coefficients were referred to the similar reactions of *n*-butanol [39] and *i*-butanol [40]. The rates of R22–R25, the H-abstraction reactions with OH radicals and O₂, were referred to the homologous reactions of *n*-butanol [38]. Besides, the rate coefficients of H-abstractions by other atoms and radicals in-

cluding O atoms, CH₃, HO₂, HCO, CH₂OH, CH₃O, and C₂H₅ radicals were also referred to as similar reactions of *n*-butanol [41]. The ipso-addition reactions of A1CH₂OH by OH radicals (R26) and H atoms (R27) could produce A1OH and A1, respectively. The rate coefficients of these reactions were obtained from Chen et al. [37]. The rate coefficients of ipso-addition reaction of A1CH₂OH by CH₃ radicals (R28) were obtained from Zhou et al. [23].

4. Results and discussion

4.1. Species identification

The PIE spectra obtained from the experimental measurements could be used to distinguish the structures of the intermediates and products. The PICs of individual species could help to distinguish isomers from PIE spectra. Figure 2 provides the PIE spectra of eleven species in the pyrolysis of A1CH₂OH, including A1CH₂ radicals, ethylbenzene (A1C₂H₅), A1CHO, acenaph-

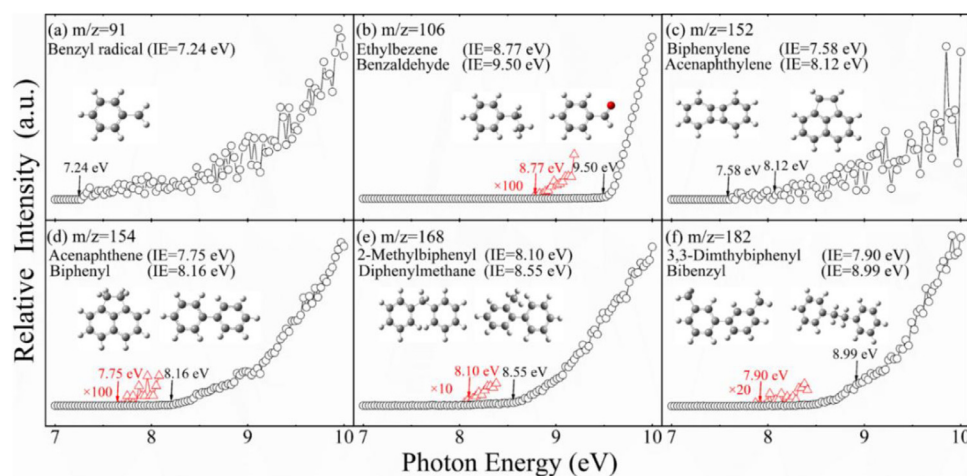


Fig. 2. PIE spectra of $m/z = 91, 106, 152, 154, 168$, and 182 in the $A1CH_2OH$ pyrolysis process at 760 torr.

Table 4

All the identified and quantified species in the pyrolysis of $A1CH_2OH$.

m/z	Formula	Species	IE/eV	30 Torr				760 Torr	
				Ref.	30 Torr	760 Torr	XM	TM/K	XM
15	CH ₃	Methyl radical	9.84 [45]	9.85	–	–	7.92E–06	1250	–
16	CH ₄	Methane	12.61 [46]	–	–	–	1.39E–05	1275	9.99E–05
26	C ₂ H ₂	Acetylene	11.4 [47]	–	–	–	1.24E–04	1275	5.62E–05
28	CO	Carbon monoxide	14.01 [48]	–	–	–	1.51E–03	1275	3.68E–03
28	C ₂ H ₄	Ethylene	10.51 [47]	–	–	–	1.80E–05	1275	2.88E–05
30	CH ₂ O	Formaldehyde	10.88 [49]	–	–	–	1.48E–04	1275	1.97E–04
39	C ₃ H ₃	Propargyl radical	8.67	8.67	–	–	2.15E–05	1275	–
66	C ₅ H ₆	1,3-Cyclopentadiene	8.57 [50]	8.58	8.56	–	9.86E–05	1275	3.06E–05
76	C ₆ H ₄	Benzyne	9.03	9.04	–	–	6.43E–06	1275	–
78	C ₆ H ₆	Benzene (A1)	9.24 [51]	9.24	9.24	–	4.31E–04	1275	1.86E–03
90	C ₇ H ₆	5-Ethenylidene-1,3-cyclopentadiene	8.29	8.31	8.30	–	2.47E–04	1275	2.90E–05
91	C ₇ H ₇	Benzyl radical (A1CH ₂)	7.24	7.25	7.25	–	2.93E–04	1275	6.59E–06
92	C ₇ H ₈	Toluene (A1CH ₃)	8.83 [44]	8.83	8.82	–	5.53E–04	1275	8.25E–04
94	C ₆ H ₆ O	Phenol (A1OH)	8.49 [50]	8.49	8.49	–	2.07E–05	1275	1.21E–05
102	C ₈ H ₆	Phenylacetylene	8.82	8.82	8.82	–	2.20E–06	1275	1.25E–05
104	C ₈ H ₈	Styrene (A1C ₂ H ₃)	8.46 [44]	8.46	8.47	–	1.41E–05	1275	2.63E–05
106	C ₈ H ₁₀	Ethylbenzene (A1C ₂ H ₅)	8.77 [44]	8.77	8.77	–	5.23E–05	1275	1.75E–05
106	C ₇ H ₆ O	Benzaldehyde (A1CHO)	9.5 [44]	9.50	9.50	–	1.01E–03	1275	1.07E–03
116	C ₉ H ₈	Indene	8.16 [44]	8.14	8.16	–	3.58E–06	1275	9.53E–06
126	C ₁₀ H ₆	1,4-Diethynylbenzene	8.58	8.67	8.67	–	4.22E–06	1275	1.35E–06
128	C ₁₀ H ₈	Naphthalene	8.14	8.17	8.20	–	3.69E–06	1275	1.07E–05
142	C ₁₁ H ₁₀	Methylnaphthalene	7.91	7.87	7.87	–	9.92E–06	1275	1.22E–05
152	C ₁₂ H ₈ A	Acenaphthylene	8.12	8.14	8.14	–	2.29E–06	1275	1.05E–05
152	C ₁₂ H ₈ B	Biphenylene	7.58	7.60	7.63	–	3.62E–07	1275	1.65E–06
154	C ₁₂ H ₁₀ A	Acenaphthene	7.75	7.76	7.76	–	5.76E–06	1275	1.20E–04
154	C ₁₂ H ₁₀ B	Biphenyl	8.16	8.13	8.11	–	6.67E–07	1275	6.29E–06
156	C ₁₂ H ₁₂	1,4-Dimethylnaphthalene	7.78	7.87	7.87	–	5.30E–06	1275	1.57E–06
166	C ₁₃ H ₁₀	Fluorene	7.91	7.84	7.84	–	1.20E–05	1275	8.67E–05
168	C ₁₃ H ₁₂	Diphenylmethane	8.55	8.53	8.53	–	7.55E–05	1275	2.74E–04
168	2-C ₁₃ H ₁₂	2-Methylbiphenyl	8.10	8.05	8.05	–	4.71E–06	1275	1.71E–05
170	C ₁₂ H ₁₀ O	Acetylnaphthalenes	8.23	8.20	8.20	–	2.29E–06	1275	8.90E–07
178	C ₁₄ H ₁₀	Phenanthrene	7.89	7.87	7.85	–	2.08E–06	1275	2.42E–05
180	C ₁₄ H ₁₂	(E)-Stilbene	7.66	7.63	7.60	–	2.76E–06	1275	8.20E–06
182	C ₁₄ H ₁₄ B	Bibenzyl	8.99	9.01	9.02	–	2.50E–05	1275	2.76E–05
182	C ₁₄ H ₁₄	3,3-Dimethylbiphenyl	7.90	7.99	7.99	–	6.25E–06	1275	6.90E–06

Note: Ref. indicates that except for those already marked in references, and other IEs come from the former database [31]; XM presents the peak mole fraction calculated against the whole experimental temperature range; TM shows the relative temperature value at peak mole fraction.

thylene ($C_{12}H_8A$), biphenylene ($C_{12}H_8B$), acenaphthene ($C_{12}H_{10}A$), biphenyl ($C_{12}H_{10}B$), diphenylmethane ($C_{13}H_{12}$), 2-methylbiphenyl ($2-C_{13}H_{12}$), bibenzyl ($C_{14}H_{14}B$), 3, 3-dimethylbiphenyl ($C_{14}H_{14}$). Figure 2a presents the PIE spectrum of $m/z = 91$, and the starting value at 7.24 eV is following the IE of $A1CH_2$ radicals [43]. Other species without isomers are identified in the same way. The PIE spectrum in Fig. 2b displays the starting values of two different signals. One initiating at 8.77 and the later at 9.50 eV, are in accordance with the IEs of $A1C_2H_5$ [44] and $A1CHO$ [44]. Similarly,

the PIE spectra of Fig. 2c-f shows the distinction of isomers with the corresponding m/z . All IEs show negligible differences with the database [31] and the errors are in a permissible range of ± 0.10 eV. Comparing the IEs and PICs tendencies of the species taken from the PIE spectra with the online data of the NIST website [43] and NSRL photoionization mass spectrometry center [31], 35 species were measured and identified (see Table 4). Compared with the oxidation of $A1CH_2OH$ [23], the pyrolysis of $A1CH_2OH$ could generate lots of PAHs, while the acids and esters produced in the

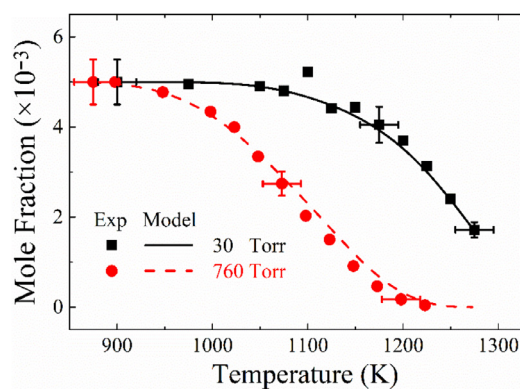


Fig. 3. Mole fraction profiles of A1CH₂OH at the pressure of 30 and 760 torr. Lines and symbols represent simulated and measured results, respectively.

A1CH₂OH oxidation were not detected which suggests the lack of oxygen in the pyrolysis process that lead to an easier aromatic species formation. The structures of the identified species with m/z larger than 75 are shown in Table S4 (SM). The mole fractions profiles of the identified species are also presented in SM. Moreover, the carbon and oxygen balance distributions confirm the reliability of the experimental data in the pyrolysis of A1CH₂OH, as displayed in Fig. S2 (SM).

4.2. Mole fraction profiles

The predicted mole fractions of A1CH₂OH and its pyrolysis products against experimental data are shown in Figs. 3–6. The current mechanism can reasonably predict the experimental data. Figure 3 shows the mole fraction profiles of A1CH₂OH in its pyrolysis at 30 and 760 torr. A1CH₂OH begins to decompose at 900 K and completely decomposes at about 1200 K at 760 torr. The onset decomposition temperature shift to 1050 K at 30 torr. The maximum temperature is 1275 K in which the conversion of A1CH₂OH is 65% at the pressure of 30 torr. Higher temperature was not reached due to the limitation of the experimental apparatus. According to the simulation tendency, it could be predicted that the A1CH₂OH will completely decompose at about 1350 K at 30 torr. In general, the current model can predict the tendencies and mole fractions profiles of A1CH₂OH well over the whole temperature range under both pressures.

The mole fraction profiles of some small species including free radicals, light hydrocarbons, and oxygenated products are shown in Fig. 4. CH₃ and C₃H₃ radicals are only observed at 30 torr due to their high reactivities. Through the ROP analysis, CH₃ radicals are mostly generated by the ipso-addition reaction of A1CH₃ with H atoms under both pressures. Phenyl radicals react with H atoms and decompose to C₃H₃ radicals. The model can predict the mole fractions and tendencies of free radicals well, which proves the applicability of the present model. Several light hydrocarbons including methane, acetylene, ethylene, and 1,3-cyclopentadiene are also shown in Fig. 4. Through the ROP analysis, methane is mostly generated by the combination of CH₃ radicals with H atoms from other species such as A1CH₂OH under both pressures. Similarly, 1,3-cyclopentadiene is mainly formed by cyclopentadienyl radicals (C₅H₅) capturing H atoms at 760 torr. However, at 30 torr, it is generated dominantly by A1OH decomposition. The main difference in the formation mechanisms result comes from differences in the experimental and simulations of 1,3-cyclopentadiene at 30 and 760 torr. Acetylene is mainly obtained through the reaction of C₃H₂ + OH = HCO + C₂H₂ at 30 torr, while it is mostly produced by the decomposition of C₁₂H₁₀A at 760 torr. The decomposition reaction of 1-phenylethyl radicals into phenyl radicals and ethy-

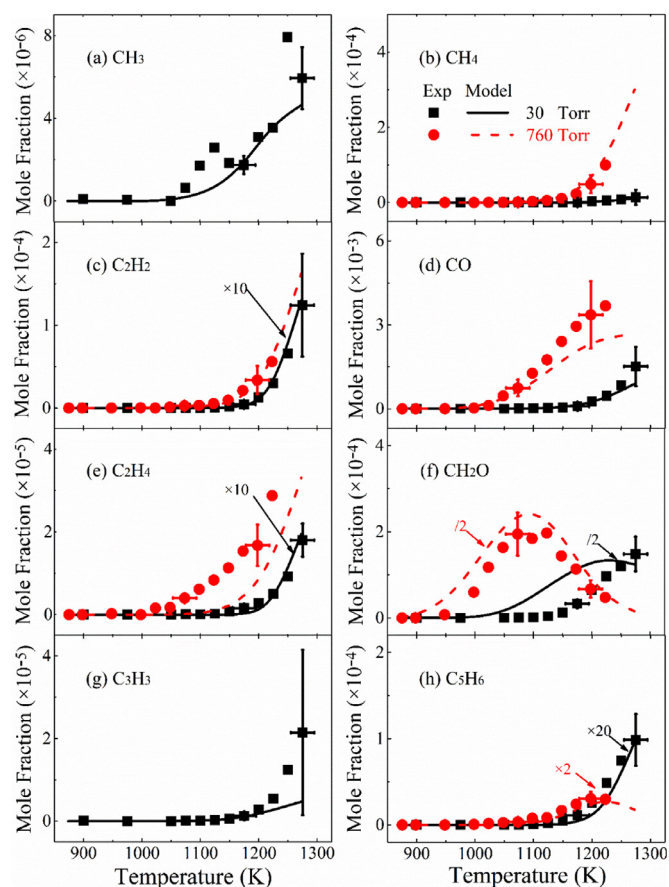


Fig. 4. Mole fraction profiles of some small species in the A1CH₂OH pyrolysis at the pressure of 30 and 760 torr. Lines and symbols represent simulated and measured results, respectively.

lene is the main route of ethylene formation under both pressures. CO and formaldehyde (CH₂O) are the major oxygenated products of A1CH₂OH pyrolysis. In addition, CO owns the highest concentration through the A1CH₂OH pyrolysis at atmospheric pressure, which is very similar to that in the former results of A1CH₂OH oxidation [23] at the ϕ of 2. CO is generated by unimolecular dissociation reactions such as A1CO = A1 + CO reaction and HCO = H + CO reaction under both pressures. CH₂O is also generated by unimolecular dissociation reactions from CH₂OH radicals. In general, the model has a good prediction for free radicals and methane, while it tends to slightly underpredict CO and other light hydrocarbons and overpredict CH₂O. From the ROP analysis, a part of CO is produced by the continuous H-abstraction reactions of CH₂O. The main routes of CH₂O consumption are H-abstraction reactions via H atoms, OH radicals, and phenyl radicals to generate HCO radicals, which will decompose into CO completely. The simulation of CH₂O is higher than the experimental value while the simulation of CO is lower than the experimental value. Therefore, this phenomenon could be attributed to the slow H-abstraction reactions rate constants of CH₂O consumption path.

The mole fraction profiles of several monophenyl ring species are displayed in Fig. 5, including benzyne (C₆H₄), A1, A1CH₂ radicals, A1CH₃, A1OH, styrene (A1C₂H₃), A1C₂H₅, A1CHO, and indene (C₉H₈). In general, the present model can predict the experimental data well. According to the ROP analysis, C₆H₄ is mainly generated by self-combination of phenyl radicals under different pressures. Some of C₆H₄ react with A1 to form C₁₂H₁₀A. There are still some discrepancies between the measured data and simulated re-

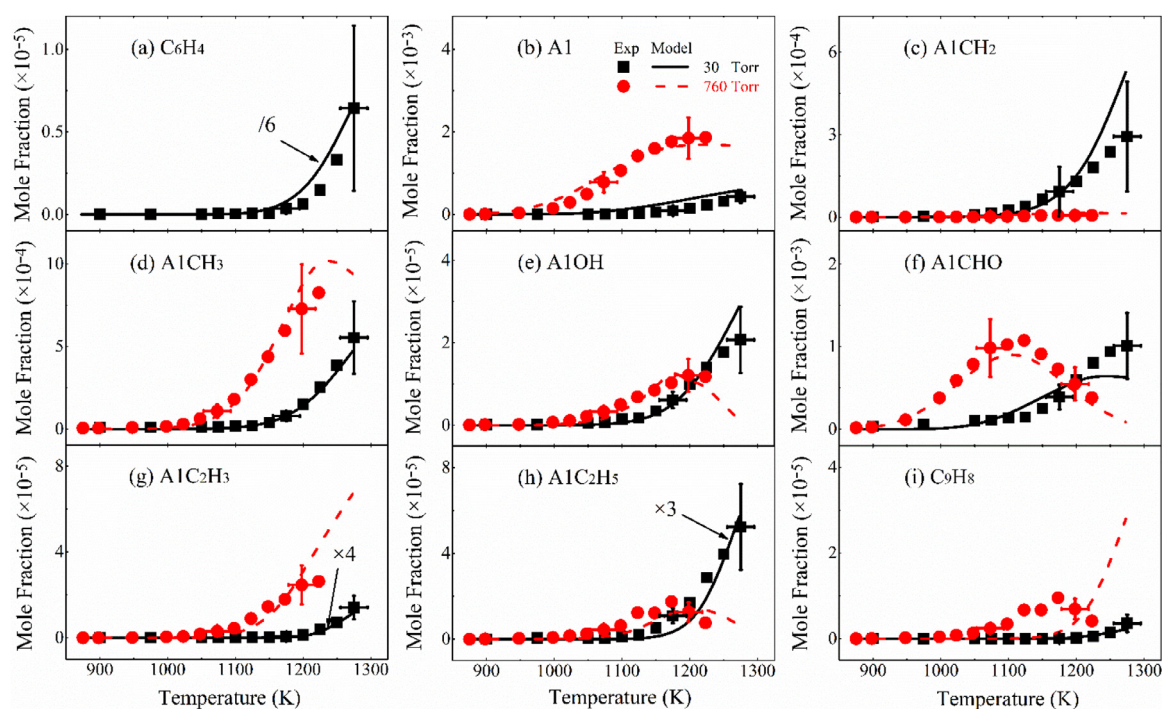


Fig. 5. Mole fraction profiles of monophenyl ring species in the A1CH₂OH pyrolysis at 30 and 760 torr. Lines and symbols represent simulated and measured results, respectively.

sults under 760 torr, as shown in Fig. 5i. Through the ROP analysis, C₉H₈ is mainly formed by the reaction A1CH₂ + C₂H₂ = C₉H₈ + H under 760 torr. Meanwhile, it is mostly consumed by the reaction H + C₉H₈ => H₂ + C₉H₇. The formation rate of C₉H₈ is much higher than the consumption rate at atmospheric pressure, which means the sub-mechanism of C₉H₈ need to be improved to increase the accuracy of the model. More detailed ROP analysis of other significant aromatic intermediates, such as A1 and A1CH₂ radicals are shown in the Section 4.3.

The mole fraction profiles of some PAHs species such as C₁₂H₁₀B, fluorene (C₁₃H₁₀) and C₁₃H₁₂ are shown in Fig. 6. The simulated results of most PAH species are in good agreement with the measured values. The concentrations of C₁₂H₁₀B and C₁₃H₁₂ are in 10⁻⁴ orders of magnitude. According to ROP analysis, C₁₂H₁₀B is primarily generated by the reaction of phenyl radicals with A1 at 30 torr, while dominantly formed by the self-combination of phenyl radicals at 760 torr. Under both pressures, C₁₃H₁₂ is mostly formed by direct addition of phenyl to A1CH₂ radicals, both being the important free radicals for the formation of PAHs. Phenyl radicals are mainly generated by A1CO radicals through unimolecular dissociation reactions. A1CH₂ radicals, obtained by direct reactions from A1CH₂OH, are generated by unimolecular initiation reactions from the cleavage of C–O bond during the pyrolysis of A1CH₂OH under different pressures. As the concentrations of A1 and A1CH₂ radicals are higher than those of other PAHs intermediates, the concentrations of C₁₂H₁₀B and C₁₃H₁₂ are in 10⁻⁴ orders of magnitude which are higher than other PAHs.

4.3. Sensitivity and ROP analysis

The sensitivity and ROP analysis of A1CH₂OH pyrolysis at conversion ratios of about 65.0% (30 torr at 1273 K) and 64.5% (760 torr at 1123 K) respectively are presented in Figs. 7 and 8. The main sixteen reactions for promoting and inhibiting A1CH₂OH consumption are shown in Fig. 7. The most significant sensitive reaction of A1CH₂OH consumption is A1CH₂ + OH => A1CH₂OH reaction under both pressures. In addition, the consumption of

A1CH₂OH is mostly promoted by the H-abstraction reactions with H atoms and OH radicals under both pressures. The addition reaction of A1CH₂ radicals and phenyl radicals, which generates C₁₃H₁₂, has a contributory influence at 30 torr while it plays an important restraining effect on the consumption of A1CH₂OH at 760 torr. Zhou et al. [23], suggested that the formation of A1CHO radicals by H-abstractions of A1CH₂OH and the reaction of H₂O₂ (+M) => OH + OH (+M) presented the important promoting influences on the A1CH₂OH oxidation consumption at both conditions. The unimolecular initiation reaction of A1CH₂OH forming A1CH₂ radicals displays the most promoting effect on the pyrolysis of A1CH₂OH while it scarcely displays the effect on the A1CH₂OH oxidation. Similarly, the reaction H₂O₂ (+M) => OH + OH (+M) shows minor effect on the A1CH₂OH pyrolysis conditions.

It is meaningful and reasonable for the comparison of sensitivity results as the conversion ratios of the A1CH₂OH consumption selected for oxidation (60% under both conditions) and pyrolysis (65.0% at 30 torr, 64.5% at 1123 K) analysis are quite close. The combination of A1CH₂ radicals with H atoms to form toluene (A1CH₃) has the most inhibiting influence on the consumption of A1CH₂OH under both pressures. Meanwhile, the H-abstractions of A1CHO with H atoms and OH radicals also show the inhibiting effects on the consumption of A1CH₂OH pyrolysis under both pressures. In Zhou et al.'s work [23], the most inhibiting reaction was the H-abstraction reactions of A1OH reacts with O₂ on the A1CH₂OH oxidation consumption at both conditions while it has little influences on the A1CH₂OH pyrolysis process. In general, the above studies suggested that the H-abstraction reactions mostly promoted A1CH₂OH consumption under both oxidation and pyrolysis conditions. However, the reactions with the most promoting effect and the most restraining effect for the A1CH₂OH pyrolysis are different from the A1CH₂OH oxidation which also have little sensitive effects for the oxidation process of A1CH₂OH consumption.

The ROP analysis of A1CH₂OH pyrolysis is presented in Fig. 8. There are four major ways for A1CH₂OH pyrolysis, namely the

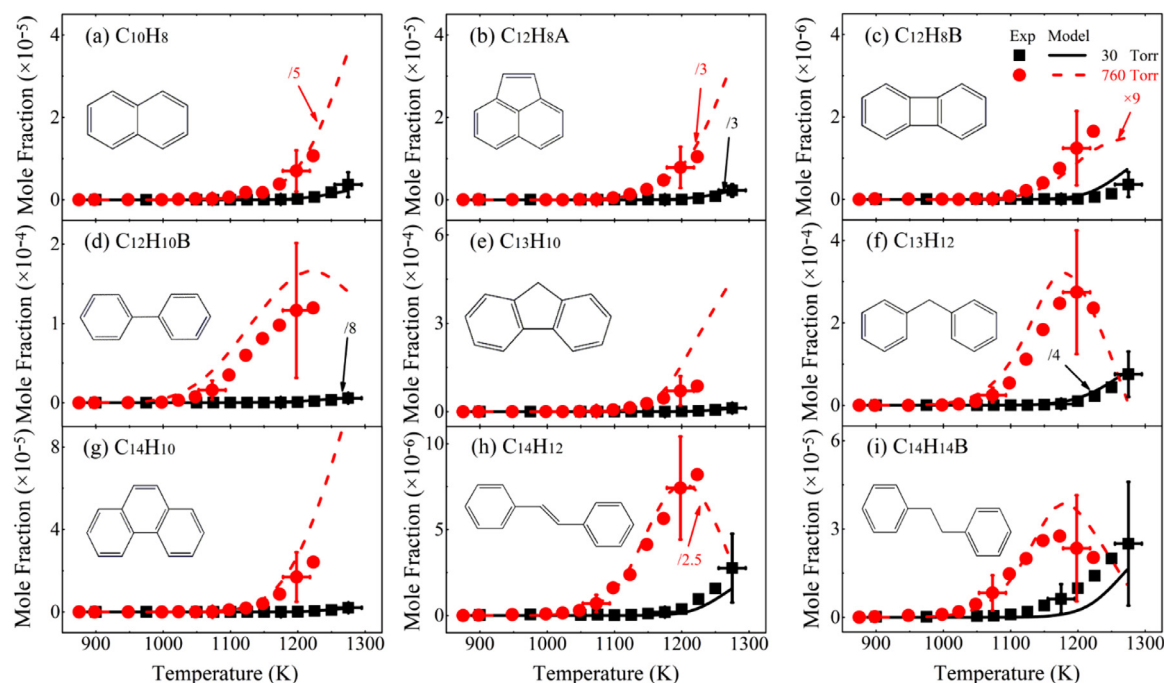


Fig. 6. Mole fraction profiles of PAHs species in the A1CH₂OH pyrolysis at the pressure of 30 and 760 torr. Lines and symbols represent simulated and measured results, respectively.

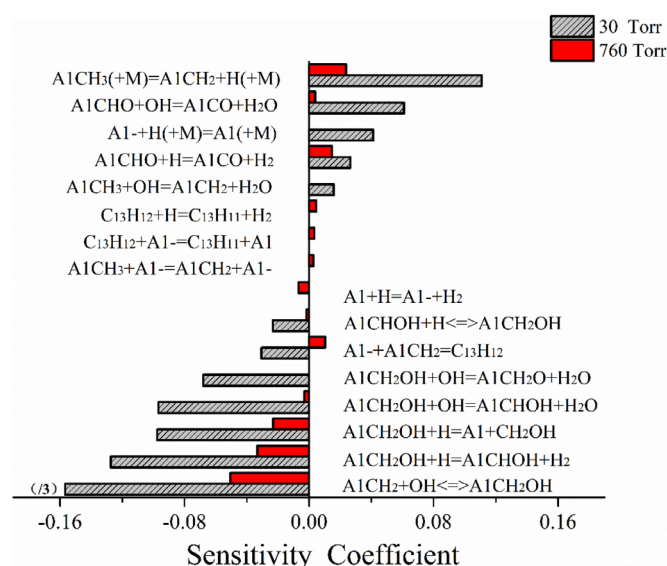


Fig. 7. Sensitivity analysis of pyrolysis of A1CH₂OH at conversion ratios of about 65.0% and 64.5% with the corresponding pressures are 30 torr (1273 K) and 760 torr (1123 K), respectively.

unimolecular initiation reaction to form A1CH₂ radicals, the ipso-addition reaction with H atoms to produce A1 and H-abstraction reactions by H atoms and OH radicals to generate A1CHOH and A1CH₂O radicals, respectively. A1CHOH radicals are mainly decomposed into A1CHO and H atoms under both pressures. A1CHOH radicals can also be consumed by isomerization to form A1CH₂O radicals, especially at 760 torr. Three consumption routes of A1CH₂O radicals are shown here, in which the major one is the unimolecular initiation reactions to form A1CHO and the other two produce are A1 and phenyl radicals. A1CO radicals, mainly generated by H-abstractions of A1CHO reacts with H atoms and OH radicals, will decompose into phenyl radicals and CO under both pres-

ures. The remaining consumption routes of A1CHO are the ipso-addition reactions that replace the aldehyde group by H atoms and OH radicals to form A1 and A1OH, respectively. Similarly, A1OH can also be produced by ipso-addition reactions of A1CH₂OH and A1 with OH radicals under both pressures. A1CH₂ radicals, phenyl radicals and A1 are the main intermediates of A1CH₂OH pyrolysis which play important roles in the formation of PAHs.

Four consumption ways of A1CH₂ radicals are related to the addition reactions, and the major channel is to react with H atoms to form A1CH₃ (55.3% at 30 torr and 46.0% at 760 torr). A1CH₃ can generate A1CH₂ radicals through H-abstraction reactions with H atoms and OH radicals (72.7% at 30 torr and 66.6% at 760 torr). The second largest proportion consumption reaction of A1CH₂ radicals is its reaction with phenyl radicals to form C₁₃H₁₂. C₁₃H₁₂ is mostly consumed through the reaction of H atoms and phenyl radicals to form C₁₃H₁₁ radicals. C₁₃H₁₀ is mainly formed by the decomposition of C₁₃H₁₁ radicals. The third one is the self-combination of A1CH₂ radicals to produce C₁₄H₁₄B. The consumption step of C₁₄H₁₄B in A1CH₂OH pyrolysis is mainly followed by C₁₄H₁₄B → C₁₄H₁₃ radicals → (E)-stilbene (C₁₄H₁₂) → A1C₂H₃ through H-abstraction, decomposition and reaction with H atoms by removing phenyl radicals, respectively. A small amount of C₁₄H₁₄B decomposes into A1C₂H₃ directly (1.2% at 30 torr and 1.8% at 760 torr). A part of C₁₄H₁₂ generates phenanthrene (C₁₄H₁₀) through H-abstraction, decomposition and isomerization reactions. The final addition reaction of A1CH₂ radicals is to react with CH₃ radicals, leads to the formation of A1C₂H₅. A1C₂H₅ can also produce A1C₂H₃ by H-abstraction and β-bond-breaking dehydrogenation reactions.

Phenyl radicals, as one of the important intermediates in the pyrolysis of A1CH₂OH, have three consumption ways. A large part of phenyl radicals (34.5% at 30 torr and 64.0% at 760 torr) react with H atoms to form A1. Most of A1 converts to phenyl radicals through the H-abstraction at 30 torr with the consumption proportion of 78.9%. Some phenyl radicals (45.8% at 30 torr and 20.8% at 760 torr) react with A1CH₂ radicals to generate C₁₃H₁₂. The consumption reactions of C₁₃H₁₂ have been discussed in the consump-

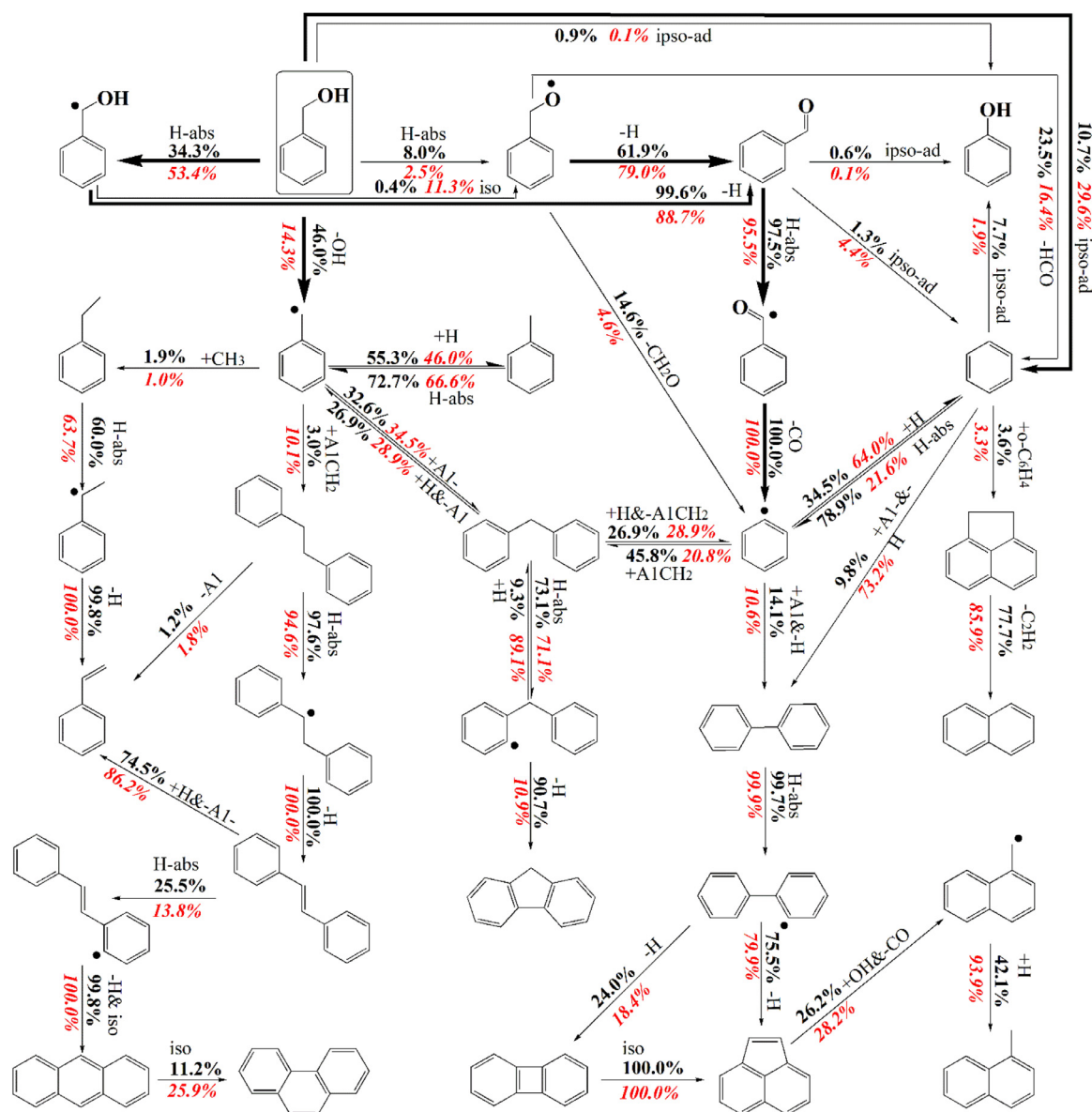


Fig. 8. Rate-of-production analysis of pyrolysis of A1CH₂OH at conversion ratios of about 65.0% and 64.5% with the corresponding pressures are 30 torr (1273 K) and 760 torr (italic, 1123 K), respectively. H-abs, iso and ipso-ad stand for the abbreviations of H-abstraction reaction, isomerization reaction and ipso-addition reaction, respectively.

tion of A1CH₂ radicals. Minor phenyl radicals (14.1% at 30 torr and 10.6% at 760 torr) convert to C₁₂H₁₀B by self-combination, which is also the main formation product of A1 (73.2% at 760 torr). Generated by H-abstraction from C₁₂H₁₀B consumption, C₁₂H₁₁ radicals are mainly consumed by decomposition reaction and transform into C₁₂H₈A (75.5% at 30 torr and 79.9% at 760 torr) and C₁₂H₈B (24.0% at 30 torr and 18.4% at 760 torr), respectively. C₁₂H₈B converts to C₁₂H₈A through isomerization reaction completely. C₁₂H₈A (26.2% at 30 torr and 28.2% at 760 torr) reacts with OH radicals, forming CO and C₁₁H₉ radicals. 1-Methylnaphthalene is mainly generated through the addition reaction of C₁₁H₉ radicals with H atoms. Most of the A1 consumption pathways have been mentioned in the above analysis. The remaining consumption reaction of A1 is A1 reacts with C₆H₄ to form C₁₂H₁₀A (3.6% at 30 torr and 3.3% at 760 torr) and it decomposes into naphthalene (C₁₀H₈) and C₂H₂.

According to ROP analysis, A1CH₂OH consumption is mostly promoted by H-abstraction reactions. The unimolecular initiation reactions of A1CH₂OH have a higher consumption proportion than

that of ipso-addition reactions at 30 torr while it shows the opposite trend at the pressure of 760 torr. H-abstractions and unimolecular dehydrogenation reactions also play important roles in the subsequent formation of PAHs. A1CH₂ radicals, phenyl radicals and A1 are the critical intermediates in the A1CH₂OH pyrolysis process, which provide various pathways for the formation of PAHs.

4.4. Validation against previous results and comparison with A1CH₂OH oxidation

To extend the validity range and confirm the accuracy of the updated A1CH₂OH mechanism, the A1CHO oxidation results from Namysl et al. [52] and Chen et al. [37] and the A1CH₂OH oxidation results of Zhou et al. [23] have been simulated. Figures S3 and S4 (see SM) present the comparison of simulation results against the A1CHO oxidation data from Namysl et al. [52] and Chen et al. [37], respectively. Both A1CHO oxidation experiments were carried out in a jet-stirred reactor under atmospheric pressure [37,52]. The

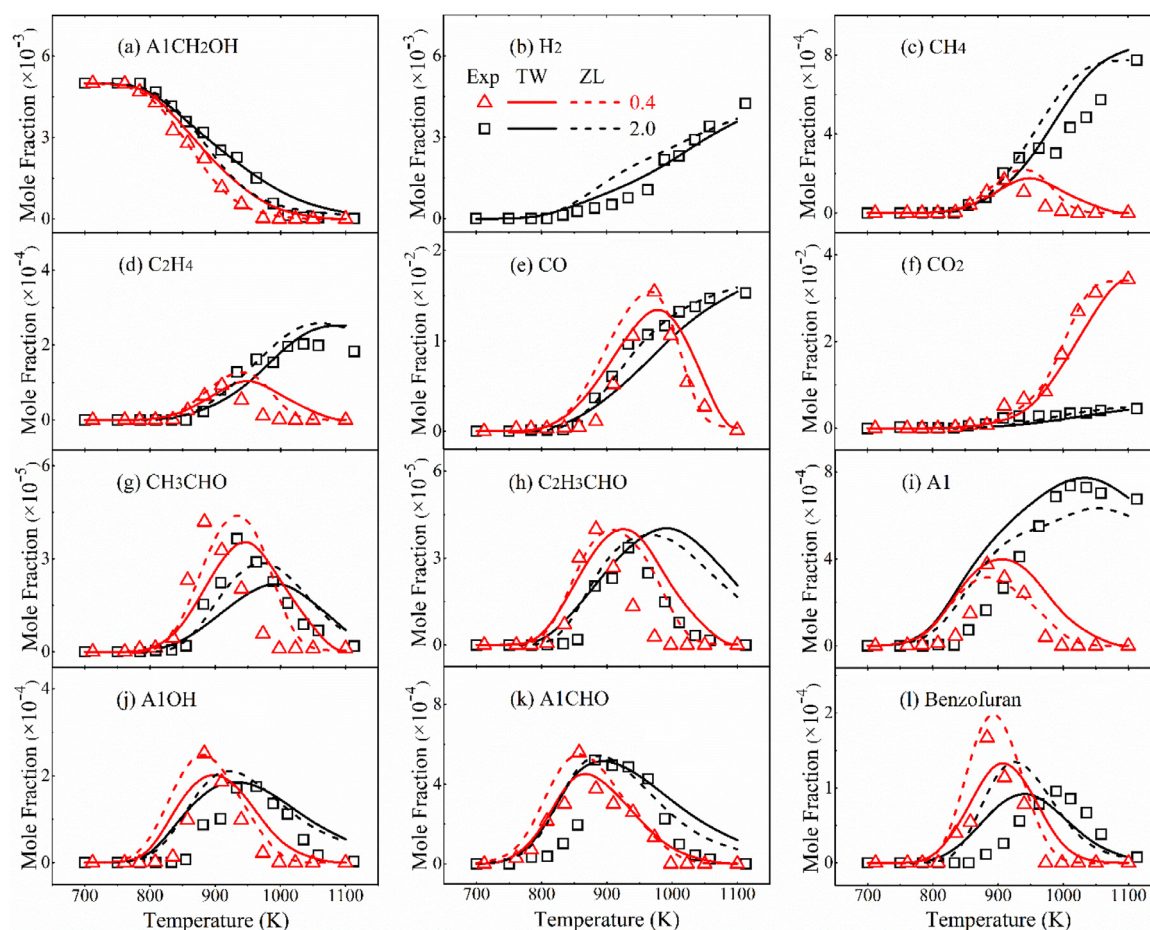


Fig. 9. The mole fraction profiles of reactant and products in the JSR oxidation of A1CH₂OH [23] under atmospheric pressure. ZL: Simulation results by using the model proposed by Zhou et al. [23]; TW: Simulation results by using the model proposed in the article.

simulated results show that the current mechanism could reasonably predict the generation of most of the intermediates and products from the A1CHO oxidation data [37,52].

Figure 9 shows the comparison among the simulated data by using the updated A1CH₂OH model and the measured A1CH₂OH oxidation data [23] and the simulated results from Zhou et al. [23]. The simulated results by the present model show a rational agreement with the A1CH₂OH oxidation experimental data. From Fig. 9, the simulated results from the updated model are slightly lower than the experimental data at the rich condition which is similar to the results from Zhou et al. [23]. This is due to the subsequent mechanism of the updated A1CH₂OH model were taken from the oxidation mechanism of A1CH₂OH [23]. However, Fig. S5 (see SM) shows that the mechanism of A1CH₂OH oxidation from Zhou et al. [23] cannot reasonably simulate the experimental results of A1CH₂OH pyrolysis, especially the formation of some radicals and monophenyl ring species. In general, there were no tremendous differences in the A1CH₂OH oxidation between the updated model and Zhou's model [23].

The updated A1CH₂OH model could also well simulate the A1CHO oxidation data [37,52], which is attributed to the fact that the updated model was mostly developed from A1CHO oxidation [37] and A1CH₂OH oxidation mechanism [23]. However, the previous oxidation mechanism of A1CH₂OH [23] could not reasonably predict the pressure effects. Therefore, the updated unimolecular initiation reactions and H-abstraction reactions of the A1CH₂OH primary mechanism extend the validity range of the updated A1CH₂OH model.

In order to further understand the effect of temperature, pressure, and ϕ on the combustion of A1CH₂OH, some results of oxidation and pyrolysis of A1CH₂OH are discussed here. A1 and A1CHO, as the main aromatic products of A1CH₂OH pyrolysis, play important roles in the subsequent aromatic formation process, which were also detected in the oxidation of A1CH₂OH [23]. The sensitivity analysis of A1CH₂OH oxidation [23] showed that the A1OH + O₂ = A1O + HO₂ reaction had the most restraining influence on the consumption of A1CH₂OH oxidation. However, the formation and consumption of A1OH rarely affect the pyrolysis process of A1CH₂OH. Figure 10 presents the comparison of the A1CH₂OH pyrolysis and oxidation results to find out the influence of temperature, pressure, and ϕ on the formation of these monophenyl ring species. The numbers above the column represent the required temperatures when the concentration reached the peak mole fractions. The oxidation experiments of A1CH₂OH were carried out at $\phi = 0.4$ and 2.0 [23] with the starting concentrations of both 0.5% which are the same as the A1CH₂OH pyrolysis experiments in this work.

With the increase of ϕ , the peak mole fractions of A1 increase while the peak mole fractions of A1OH decrease under the same pressure. A1 is mostly generated by the ipso-addition reaction, in which A1CH₂OH reacts with H atoms. There are more H atoms in rich oxidation which lead to a higher forming rate for the production of A1. Due to the decrease of reaction activity, the pyrolysis peak mole fraction of A1 at the pressure of 30 torr is lower than that at 760 torr. However, A1 can also be formed through phenyl radicals by capturing H atoms from other

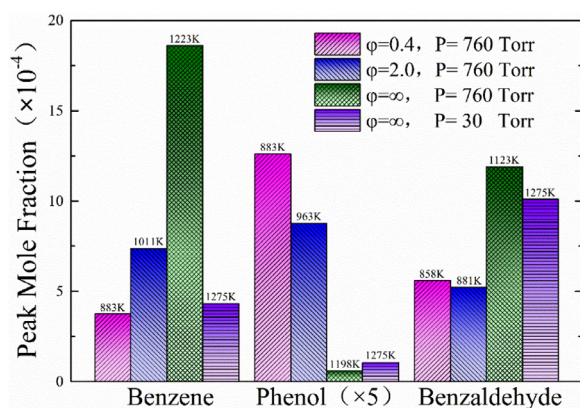


Fig. 10. Comparison of monophenyl ring species between pyrolysis and oxidation of A1CH₂OH.

species, which leads to the peak mole fraction at 30 torr which is slightly larger than that at $\phi=0.4$ and atmospheric pressure. A1OH is mostly formed through the H- abstraction reaction, A1O radicals react with HO₂ radicals at both conditions while the main formation reaction will turn into the ipso-addition reaction between OH radicals and A1CH₂OH with the decrease of oxygen concentration. Due to HO₂ radicals are mainly generated by O₂ capturing H atoms from other species such as HCO radicals, the concentration of HO₂ radicals will decrease when the concentration of O₂ decreases, which leads to the change of the main generation route of A1OH. The formation and consumption of A1OH are closely related to the change of O₂ concentration at atmospheric pressure. Meanwhile, the oxidation process of A1CH₂OH is greatly affected by O₂ concentration, which is the reason that the reaction A1OH + O₂ = A1O + HO₂ has the most inhibiting influence on the consumption of A1CH₂OH oxidation under atmospheric pressure. Furthermore, A1OH is mainly generated by ipso-addition reactions between OH radicals and other aromatic species like A1CH₂OH at the pyrolysis process. With the decrease of pressure, the proportion of unimolecular initiation reactions from A1CH₂OH to form A1CH₂ and OH radicals shift to a higher ratio, results in a higher concentration of OH radicals. Therefore, the peak concentration of A1OH at 30 torr is higher than that at 760 torr. Sensitivity analysis shows that the reactions from A1OH rarely influence the A1CH₂OH pyrolysis because most OH radicals are consumed by the H-abstraction reactions to form H₂O instead of the ipso-addition reactions to generate A1OH.

The formation of A1CHO is relatively stable with various ϕ and pressure. A1CHO is mostly formed by dehydrogenation reactions from A1CHOH radicals that generated through the H-abstraction reactions of A1CH₂OH. Under different conditions, A1CHO is consumed by the H-abstraction reaction and decarbonylation reaction to form phenyl radicals. Phenyl radicals tend to form PAHs under pyrolysis conditions while reacting with O₂ to form A1OO radicals and decompose into small molecules. This is one of the main reasons that more PAHs can be observed during the pyrolysis process when compared with oxidation.

5. Conclusion

This work reports the experimental and the modeling studies of A1CH₂OH pyrolysis with synchrotron radiation photoionization and molecular-beam mass spectrometry at 30 and 760 torr over the temperature range from 873 to 1275 K. The accuracy of the experimental data is confirmed by the carbon balance and oxygen balance calculated from the mole fractions of 35 measured and quantified species. The updated mechanism provided in this

work could not only predict the pyrolysis experimental data of A1CH₂OH well but also reasonably against the experimental data of A1CH₂OH oxidation [23] and A1CHO oxidation [37,52]. Sensitivity and ROP analysis are presented to give further understanding of the combustion chemistry of A1CH₂OH. Under both pressures, the reaction A1CH₂ + OH $\rightleftharpoons$ A1CH₂OH is the most promoting sensitive reaction for the A1CH₂OH pyrolysis while the A1CH₃(+M) = A1CH₂ + H(+M) reaction has a significant restraining effect. Comparing with A1CH₂OH oxidation data, the peak mole fractions of A1 increase under the same pressure while the peak mole fractions of A1OH decrease with the increase of ϕ . Due to the decrease of reaction activity, the pyrolysis peak mole fraction of A1 at the pressure of 30 torr is lower than that at 760 torr. The formation reaction of A1OH at low pressure is changed compared with the atmospheric pressure. However, the formation of A1CHO is relatively stable with various ϕ and pressure. These results could extend the pool production of A1CH₂OH pyrolysis and provide deep insight into the combustion characteristic of A1CH₂OH.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2021.111477](https://doi.org/10.1016/j.combustflame.2021.111477).

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